

Scalability & Future Plan

Date	1 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application is designed for a limited number of concurrent users.	Performance may decrease if many users access the system simultaneously.	High
2	The model is trained on a single historical dataset.	Prediction accuracy may decrease for new or changing applicant trends.	High
3	The system predicts only approval or rejection without providing detailed reasons or recommendations.	Users and analysts receive limited decision insights	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users	Deploy on a scalable cloud platform and use load balancing to support multiple concurrent users.
2	Data Storage	Uses a static training dataset.	Continuously update the dataset with new applicant records and retrain the model periodically.
3	Performance	Predictions are generated using a single deployed model.	Optimize preprocessing, improve model efficiency, and implement caching for faster response times.

4	Security	Basic input validation and application security.	Add secure authentication, HTTPS, role-based access control, sensitive applicant information.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve prediction accuracy by training on larger and more diverse datasets.	Week(1-2) days	Better prediction performance and reduced false approvals/rejections.
Phase 2	Implement user authentication and prediction history for applicants and administrators.	Week(3-4) days	Improved security, user management, and record tracking.
Phase 3	Integrate explainable AI techniques to provide reasons for approval or rejection decisions	Week(5-6) days	Increased transparency and trust in model predictions.
Phase 4	Deploy the application on a cloud platform with real-time monitoring and support for concurrent users.	7 days	Improved scalability, availability, and production readiness.